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Level-1 Casing-While-Drilling Technology Utilization Across Extended Drilled Open-Hole Sections Reduced Operational Time and Ensured Operations Continuity

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Abstract

In extended open-hole sections, the industry's typical operational sequence to preserve wellbore cleaning is performing frequent wiper trips to remove ledges/tight sections (example of the frequency is every 2,000 ft drilled) or performing single wiper trip across the entire open-hole. To reduce the operational time and operational risks of tripping in/out of the extended drilled section, level-1 casing-while-drilling (CwD) is utilized to hard ream the casing string across the entire open-hole to section total depth. It also serves to optimize the casing design of different areas to interest by combining multiple trouble zones in one section.

To ease running casing through the extended already drilled open-hole, single or several wiper trips are performed clear ledge/obstructions/tight holes if any. After reaching the previous casing, the drill string is run back through the drilled section, reaming through obstructions/tight holes. The drill string is pulled out again to surface. This operational sequence consumes about 2.5-3 days, in addition to the increasing risk of getting the string stuck. All these operational sequences are replaced by the utilization of the Level-1 CwD technology to hard ream the casing string. The operational days and associated extra cost are controlled and reduced.

Main challenge in drilling through unconsolidated sand zones in shallow sections, and time-dependent unstable shale zones in deeper sections is inability to secure them without significant rig time delay. Loss circulation zones exist around these problematic zones. When severe to complete loss of circulation is experienced, the complexity of securing the problematic zones increases. The level-I casing-while-drilling (CwD) technology was utilized successfully to hard ream and permanently isolate the problematic zones without nonproductive time.

CwD were successfully utilized to hard ream the casing strings to target depth to case-off the severely unstable zones experienced in the offsets. The additional NPT trips were eliminated completely. In addition, the 9-5/8-in casing string hard reaming was performed while incorporating a multistage differential valve (DV) and cemented in two stages. During the job execution, the reaming parameters are constantly monitored to not exceed casing strings limitations, especially the accumulated fatigue on the string as they are rotated. As a result, the troubles experienced in offset wells were successfully managed.

Complexity in drilling wells is increasing. Therefore, introducing innovative solutions that enable reducing the number of extra trips, is essential to reducing NPT. The offset wells drilling troubles related to securing unstable formations were studied thoroughly. Level-I CwD technology was successfully utilized to eliminate those identified troubles without adding additional time-delaying trips. The unstable zones were secured in a record time compared to the offsets. The results were reaming the casing through extended open-hole sections until reaching the longest open-hole interval worldwide to be reamed with the 9-5/8-in casing allowing combining and isolating multiple trouble zones together.

Introduction

CwD technology, the logic used to arrive at these solutions, and the results of implementation will be discussed in detail in this paper and future considerations will be reviewed.

Casing while drilling is a relatively new well construction process in which standard casing is used as a drill string to deliver both rotary & hydraulic power to a drill bit and/or BHA connected to the casing. Attempts to use CwD were reported in 1920 (Dipal Patel, 2019). Since CwD technology was introduced for field trials in 1999 (Warren, et al., 2001) it has gradually matured to address various drilling problems and challenges to the extent it is now used in many applications to drill from one casing shoe to the next casing point in vertical, tangent & directional sections.

Maturity of the technology has led to multiple casing drilling technology configurations being available for field implementation (Fig. 1) IPTC-19730-MS as highlighted below:

- i. Non-retrievable system.
- ii. Retrievable BHA system.

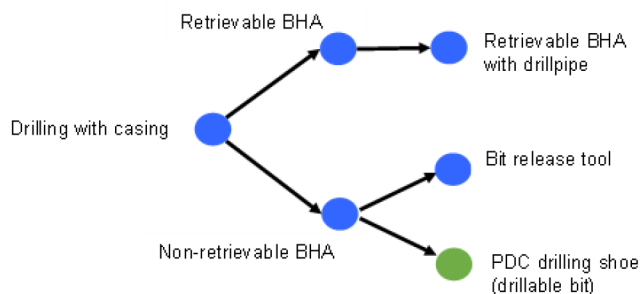


Figure 1—CwD techniques (IPTC-19730-MS)

Non-Retrievable System uses a rotating tool to turn the casing at surface and a drillable bit made up to the casing on bottom to drill the formation, the casing in this system is considered as the drill string. Accessories are installed along the casing string to limit vibration and centralize the string for cementing after reaching Target Depth (TD). This system can be implemented in any size casing. Drilling is typically performed in a single run and does not allow the use of directional or logging tools.

Retrievable BHA System employs the same casing rotating tool and accessories as the non-retrievable BHA system but replaces the drillable bit with a retrievable BHA. While the casing remains to be the drill string, the retrievable system allows for the inclusion of conventional drilling bottom hole assembly components, such as a mud motor, RSS, and/or MWD. The retrievable system allows retrieval of the BHA through the drift of the casing at any point in the drilling process using standard drillpipe and specialized retrieval tools.

The deployment of CwD allows for the casing to be constantly on bottom, reducing the potential for incidents causing non-productive time (NPT) due to wellbore instability. Also, the technology allows for further casing design optimization. CwD delivers a superior wellbore by casing off and stabilizing the

formation while drilling. Each foot of borehole is drilled and cased off simultaneously eliminating the need for a separate casing run.

CwD System Selection

CwD system(s) has widely expanded since its introduction as a new commercial technology in 1999. The systems divided into 2 different types with implementation in multiple locations around the world, as shown in (Fig.2) below.

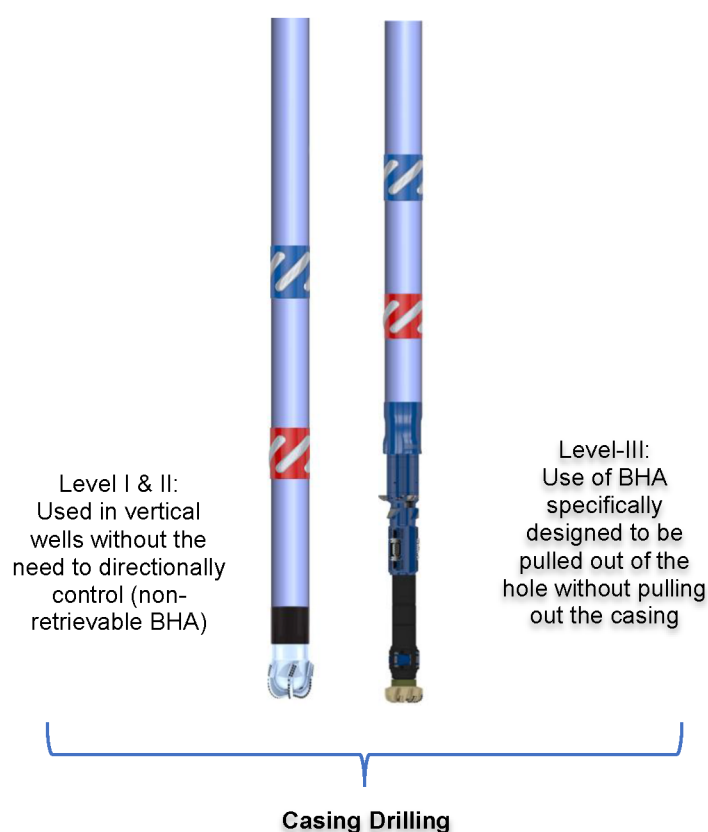


Figure 2—Casing-while Drilling Levels (IPTC-19730-MS)

For optimization project, Level 2 system requires a special drill shoe which is combination of drillable alloy body, fixed cutter bit profile and Polycrystalline Diamond Compact (PDC) cutters. Having these features allow the special drill shoe to be drilled out with a standard PDC bit without the need of dedicated clean-out trip, improving drilling performance and effective hole cleaning.

Requirement for CwD Technology

CwD is proven technology to provide solution where total losses and/or borehole instability is registered as a critical drilling challenge towards successful well delivery.

Solutions CwD introduces towards addressing these challenges are briefly described below:

1. Challenge: NPT caused by loss circulation and poor wellbore instability.
 - a. Often reduces losses.
 - b. CwD provides the ability to drill "blind" with no or partial returns.
 - c. Better control on mud cap and related chemicals.
2. Challenge: Poor wellbore stability in highly depleted formations/reservoirs or interbedded sections

- a. Plastering effect strengthening the borehole wall (smearing cuttings on the borehole wall) while rotating the casing during drilling. The smearing action strengthens the wellbore while it seals pores in the formation to reduce fluid loss (Fig.3)
 - b. Assurance of casing setting at desired depth
 - c. Casing string elimination if the casing string is used for hole problems.
3. Challenge: Complex casing designs and time savings (faster process)
 - a. Ability to slim well designs through Casing Drilling: While casing drilling, since we have the ability to case-off the well while drilling, in some applications we are able to eliminate a casing string entirely by gaining the ability to set the parent casing deeper than originally planned.
 4. Challenge: Optimizing time and reducing well delivery cost(s)
 - a. Hole cleaning improvement: CwD introduces significantly higher annular velocities, a critical factor in hole cleaning, than conventional drilling with the same flow rates.
 5. Challenge: Personnel and environmental safety.
 - a. Fewer people used with CwD operations compared to conventional operations reducing exposure to health and environmental incidents.

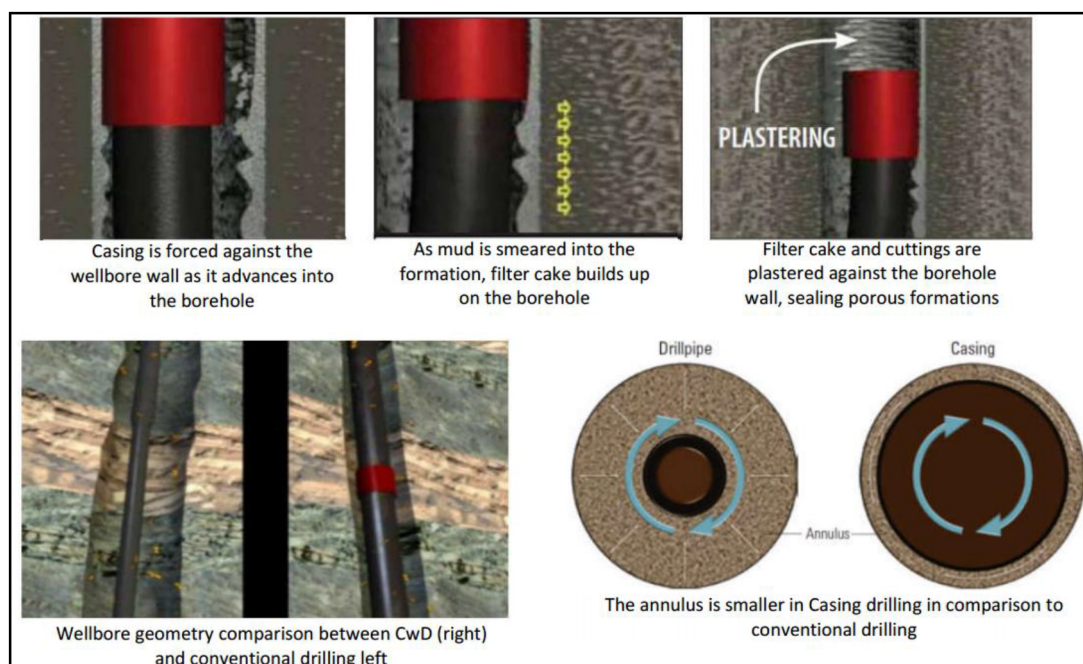


Figure 3—Plastering effect and wellbore geometry on CwD (SPE 147102)

In the case being covered, CwD was introduced to address challenges related to wellbore instability caused by total losses in the target hole section. In the direct offset well, total fluid losses led to poor wellbore stability. Due to the severity of wellbore instability, the operator had performed cased hole & open hole side-track respectively, after many days of remedial operations a decision was made to introduce CwD as a solution to mitigate the drilling challenges & ensure section is cased and secured.

Job Planning: 9-5/8-in & 13-3/8-in

Subject wells were planned as an unconventional development horizontal water disposal well. Offset well analysis has shown particular wellbore instability across 9-5/8-in & 13-3/8-in sections related to drilling

through sticky shales and loose sandstone zones charged with downhole faults/fractures to lost circulation zone. Loss circulation is commonly experienced which increases the complexity of controlling wellbore collapse. It results in stuck pipe incidents, inability to run casing to bottom, etc. In seldom incidents drilling or tripping through unstable zones with complete loss of circulation, severe tight spots, stalling tendencies, hard back-reaming, etc. might be experienced. In the worst case, the pipe might get stuck or had to pull out casing as it could not reach to bottom with conventional way, The engineering team investigated several options allowing casing off the trouble zone and the solution was to utilize level-I casing-while-drilling technology to ream and case-off the instable zones. Extensive engineering simulations, technical limits, and risks assessment are set insuring flawless execution.

The main challenges of the section were (Fig.4):

- Avoid stuck pipe incident by proper management of drilling practices under total losses.
- Formation instability due to loss circulation zones in addition to centralizers lost in hole which increases complexity.
- Ensure casing reached to bottom & isolate troublesome formation.
- Formation was exposed for long time due to inability to run casing conventional.

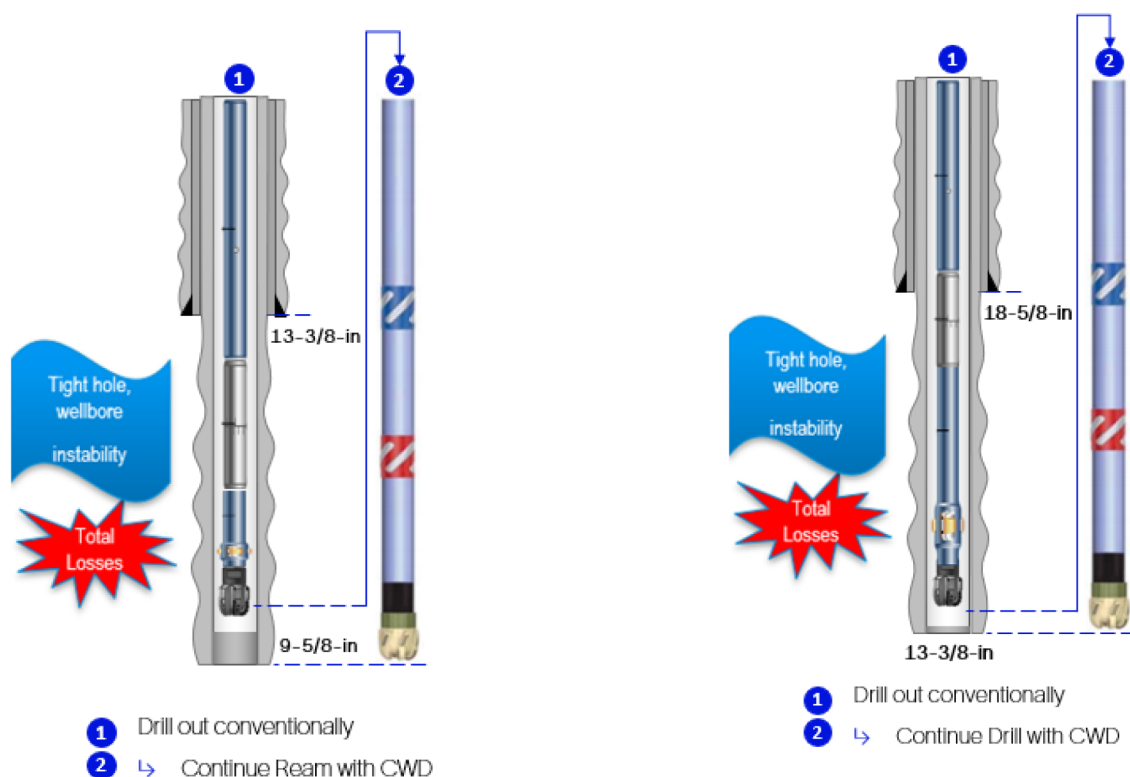


Figure 4—Well-Schematic

Engineering Simulation

As part of the planning to perform engineering simulation to confirm that applying casing drilling is technically feasible and that the torque & drag, stress & fatigue analysis are within the technical acceptable range. The fatigue analysis with the actual survey was acceptable however torque range had shown an alarming result due to total losses & expected junk in hole (Fig.5) as such close monitoring of the torque during reaming is critical to ensure no limits were crossed which could lead to casing connection fatigue and twist-off.

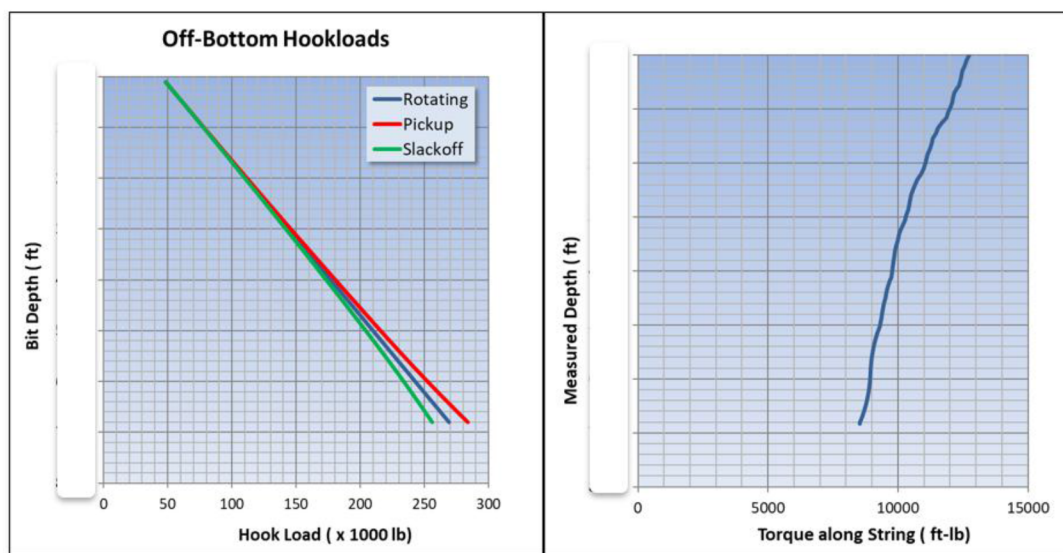


Figure 5—Planned Torque & Drag Graphs Simulation

Execution & Results 9-5/8-in

Short Open-hole Section

The integration of the extensive engineering simulations to set the technical limits, drilling fluids strategy, and proper tools utilization (centralizers, cementing diverter valve tool) led to a marked improvement in the level-1 CwD operational performance. During the hard reaming of the casing job execution, the reaming performance is constantly monitored and cross-checked with the initial simulations to verify not exceeding the technical limits of the surface and downhole tubulars (casing string, casing running tool, drill shoe, diverter valve tool, etc) or the drilled formations (not exceeding the fracture gradient). The drilling fluid system is converted to optimal casing running type to improve the casing hard-reaming performance by increasing the lubricity and friction reduction. This strategic approach not only eliminates adding additional operational days to the wells' duration, but also significantly minimized drilling risks associated with the additional tripping in/out of the extended drilled section. No significant associated operational costs are added. Furthermore, global records of hard reaming casing through extended drilled open-hole sections are set.

16-in & 12-1/4-in sections were planned to be drilled conventionally to casing point in two different areas of interest. Then the drilling assembly is pulled out. The 13-3/8-in and 9-5/8-in casing is ran through the drilled wellbore with level-I casing drilling to ream casing to bottom, to pass through tight spots or even re-drill collapsed formation zones without doing a reaming trip with the drilling assembly.

The level-I casing drilling BHA (Fig.6) was then picked up with 9-5/8-in BTC casing connection equipped with torque-rings that provides the torque capabilities required during reaming with efficient seal ability that was anticipated as per engineering T&D simulation. An appropriate float collar was selected to ensure it can withstand the high flow rates required for reaming and hole cleaning without compromising valve integrity during the cementing operations. Two float collars were used as a precaution.

During job execution, the drilling performances were constantly monitored. The engineering simulations were updated using the actual parameters to ensure accurate measurements of the accumulated fatigue while being rotated to preserve the casing. Furthermore, the hydraulics are optimized real-time to ensure hole cleaning without further increase in the equivalent circulating density.

CASING DRILLING SHOE TRACK									
PU Order	COMPONENTS	OD in	ID in	M/U Torque with MLT-TQ (F.Lbs)			Weight ppf	Connection (Top/Bottom)	
				Min	Opt.	Max.		Type	Gender
G	CASING (All the rest to the Surface)	9 5/8	8.75	28,000			43.50	BTC BTC	Box Pin
F	CASING JOINT	9 5/8	8.75	28,000			43.50	BTC BTC	Box Pin
E	Float Collar	9 5/8	8.75	28,000			43.50	BTC BTC	Box Pin
D	CASING JOINT	9 5/8	8.75	28,000			43.50	BTC BTC	Box Pin
C	Float Collar	9 5/8	8.75	28,000			43.50	BTC BTC	Box Pin
B	CASING Joint	9 5/8	8.75	28,000			43.50	BTC BTC	Box Pin
A	9-5/8" XCD/R	12.00	8.75	16,000				BTC	Box

Note: XCD Connection Baker-Locked

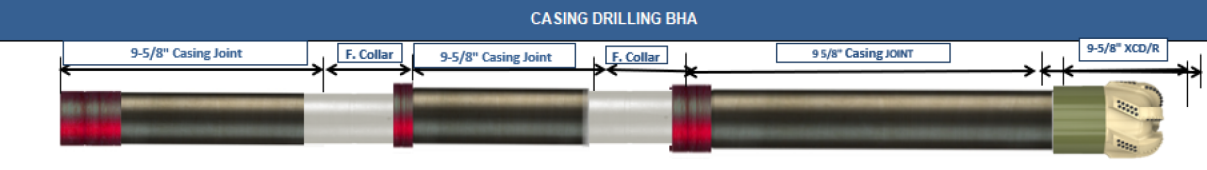


Figure 6—9-5/8-in level I casing-while-drilling (CwD) BHA

Accumulated fatigue and torque and drag (T&D) simulations were performed during the job execution to ensure the total revolutions do not exceed the maximum value to cause fatigue, and torque within the acceptable technical limits for all string components.

As a result, the CwD operation went as planned, with no registered incidents. Successfully reamed to section TD covering troublesome formations with parameters optimization applied as per (Fig.8) Achievements & highlights:

- No safety related incidents.
- CwD reamed 12-1/4-in Section
- Utilize 9-5/8-in BTC casing with torque-rings (Fig.7) rings to enhance torque capabilities.
- Optimize hydraulics to ensure hole cleaning without increasing ECD to induce losses
- Save the well from side-track & additional NPT.



Figure 7—BTC MLT-TQ Ring Specs

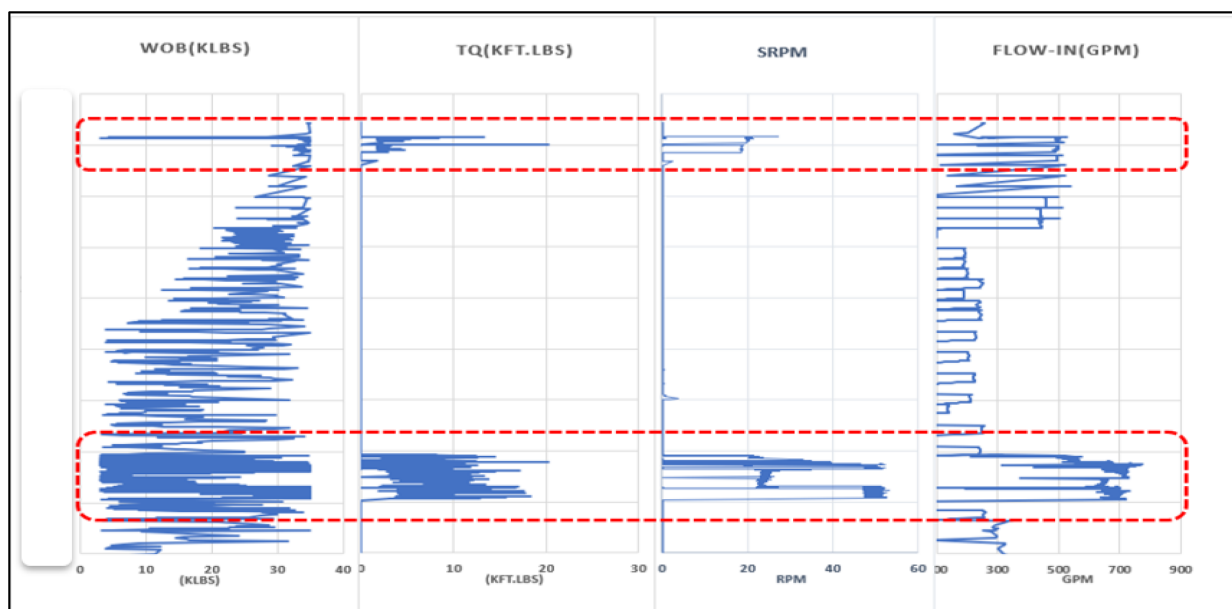


Figure 8—9-5/8-in Level-1 Performance Analysis for the short open-hole section.

Long Open-hole Section (Longest already drilled open-hole section with Level-1 CwD Application Worldwide)

In a very challenging case, drilling through 16-in section with total losses to expected depth which had to be shortened due to stuck incidents and hole condition were deteriorating. Applied hard back-reaming to be able to pull-out and decided to eliminate wiper trip and deploy 13-3/8-in Level-1 CwD reaming application to ensure casing to bottom. The 13-3/8-in application for this well will be illustrated after highlighting the 9-5/8-in casing run. The 12-in section now becomes very challenging as the planned total is extended now, forcing drilling through different layers with different drilling troubles for each such as potential losses and sticky reactive shale. As the 13-3/8-in casing was set 600-ft shallower than, this led to an increase in the 12-in open hole interval which increases the well complexity causing wellbore instability.

The level-1 CwD BHA has a diverter valve (DV) assembly added in the casing string to be able to perform a two-stage cementing operation, which adds to the complexity of the execution. The DV assembly has to be centralized with special centralizers installed above and below it. This is done to protect the DV assembly

from severally and continuously contacting the previously installed 13-3/8-in casing string, resulting in twisting off the casing string (Fig.9).

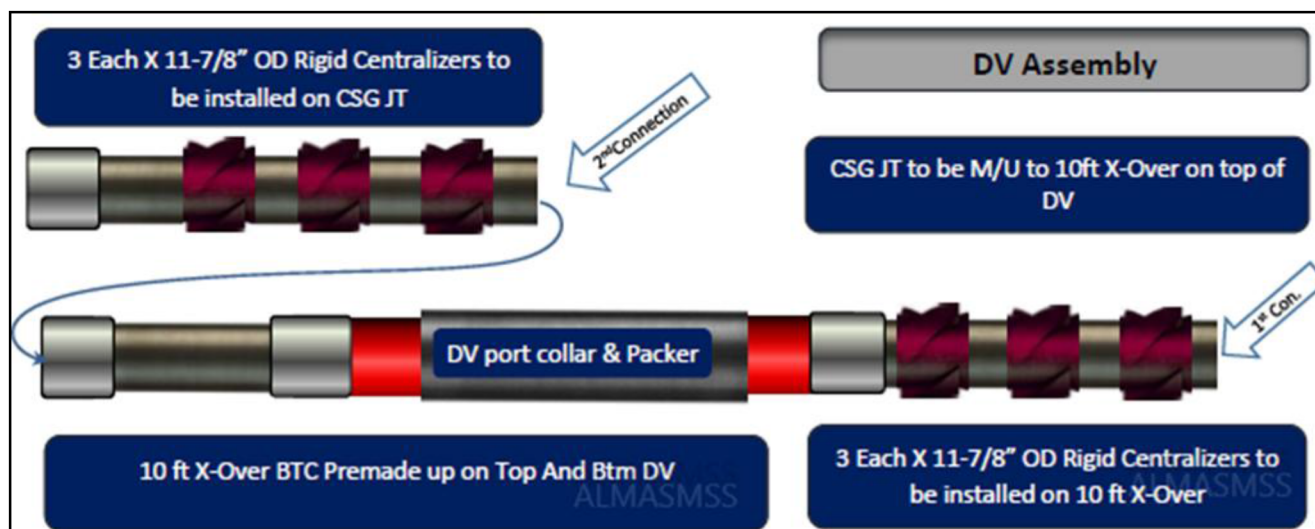


Figure 9—Diverter Valve Assembly Installation and Centralization Plan for the long 9-5/8-in open-hole section.

Accumulated fatigue and torque and drag (T&D) simulations were performed (Fig.10). Since the wellbore was mostly vertical, the accumulated fatigue simulation resulted in a maximum total revolutions of almost 240,000 RPM.

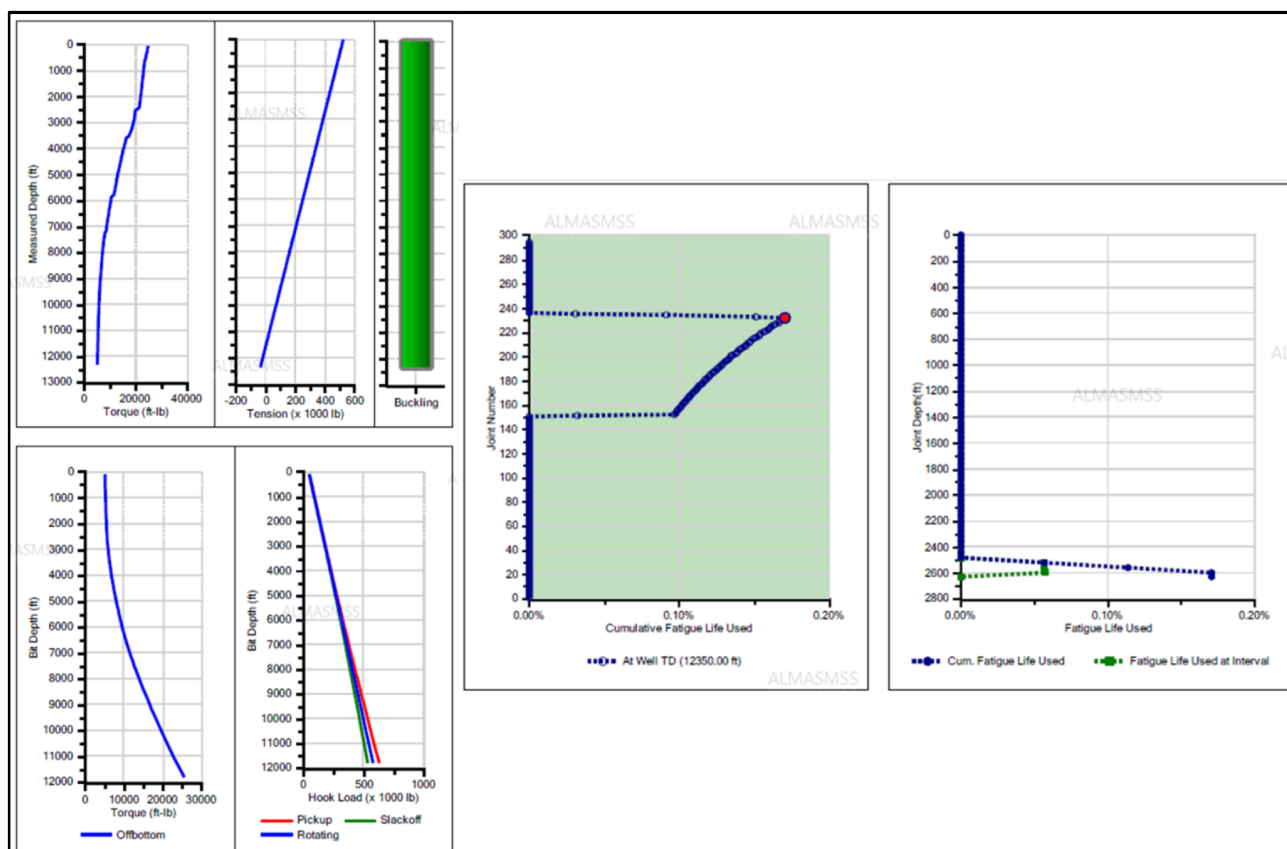


Figure 10—9-5/8-in Level-1 Performance Analysis for the short open-hole section.

The summary of the longstring open-hole worldwide application for level-1 CwD is summarized in [figure 12 \(Fig.12\)](#). The chart above indicates that the casing went through three hard reaming intervals. Zooming in into one of the hard reaming intervals in the second chart, the average surface rotary speed starting with 20 RPM and then increased to 40 RPM to be able to pass through the interval. The weight-on-bit applied (WOB) had to be increased from 10,000 lbs to 30 lbs. The surface torque reached the maximum torque set at surface at 20,000 ft-lbs at some intervals. This is higher than the planned parameters shown in [\(Fig.11\)](#). The actual surface rotary speed, WOB, and even the surface torque limit had to be in crease more than planned to be able to pass the casing string through this very tough zone. The actual total revulsions before the casing gets fatigued were almost reached. Clearly, the casing string will not be run conventionally by washing the casing string down through the hard reaming intervals.

Hole size	Formation	Footage	RPM	WOB(Klbs)	GPM	Max OPS TQ (ft-lb)	Comments
(in)		(Ft)					
12			20-30	5-10	500-550	18,000	in case low progress stage parameters as per table. Monitor hole cleaning trend, Optimize RPM and monitor T&D trend.

Figure 11—Reaming Planned Parameters for the long 9-5/8-in open-hole section.

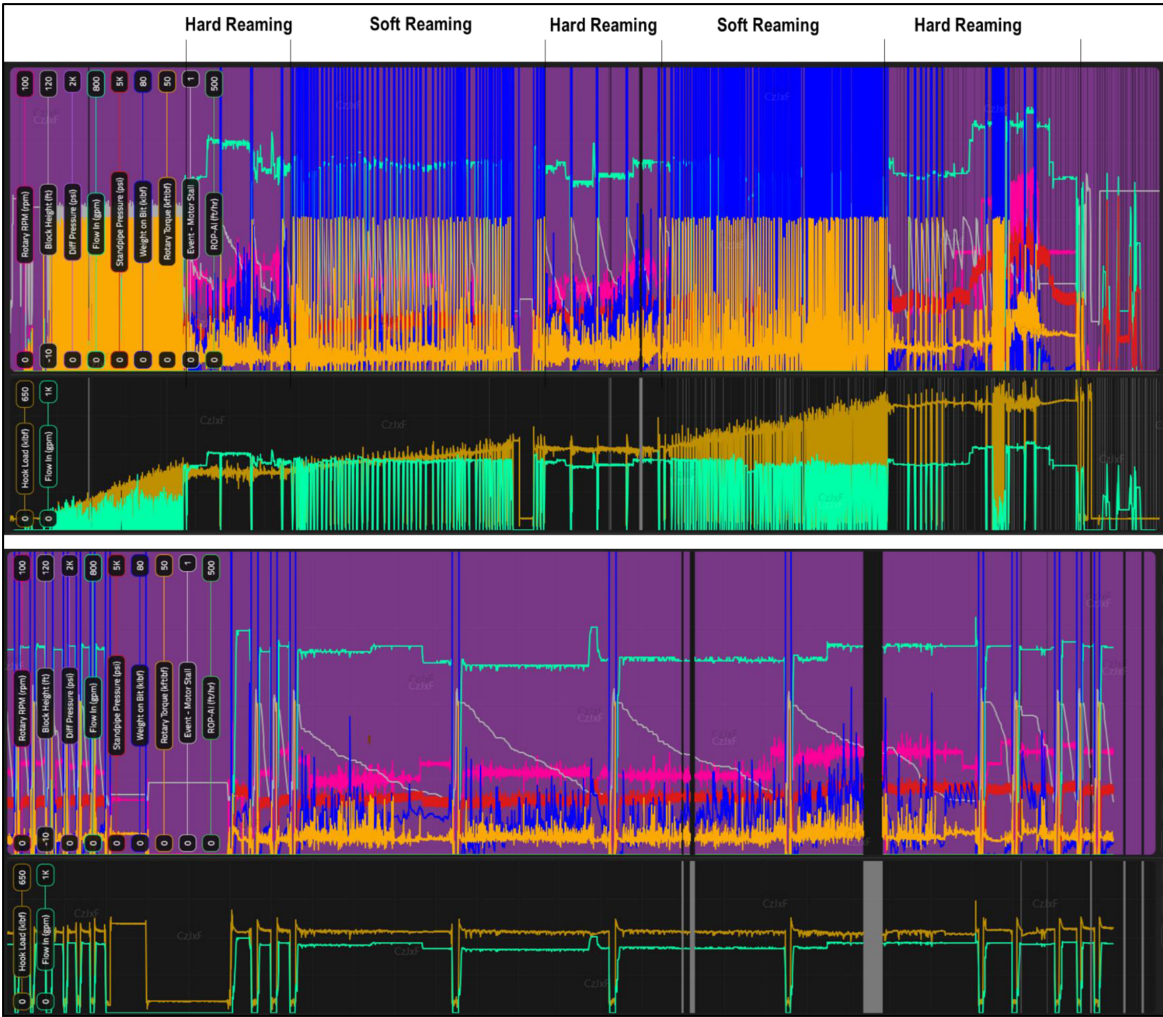


Figure 12—9-5/8-in Level-1 Performance Analysis for the longest level-1 CwD open-hole Application Worldwide. The above graph shows the total job duration. The bottom chart illustrates a zoomed-in section of one of the three hard reaming intervals.

Execution & Results 13-3/8-in

CWD Level-I technology as a risk mitigation in the troublesome 16-in section as per offset wells based on losses rate and wellbore conditions during POOH. Proper engineering, T&D (Fig.13) and hydraulics simulation were prepared to identify operating parameters window & technical limits. While drilling conventionally encountered total losses & encountered tight spots during POOH trip, eliminate wiper trips & deploy CWD Level-I where had to wash down open hole & apply reaming parameters with BHA configuration as (Fig.14).

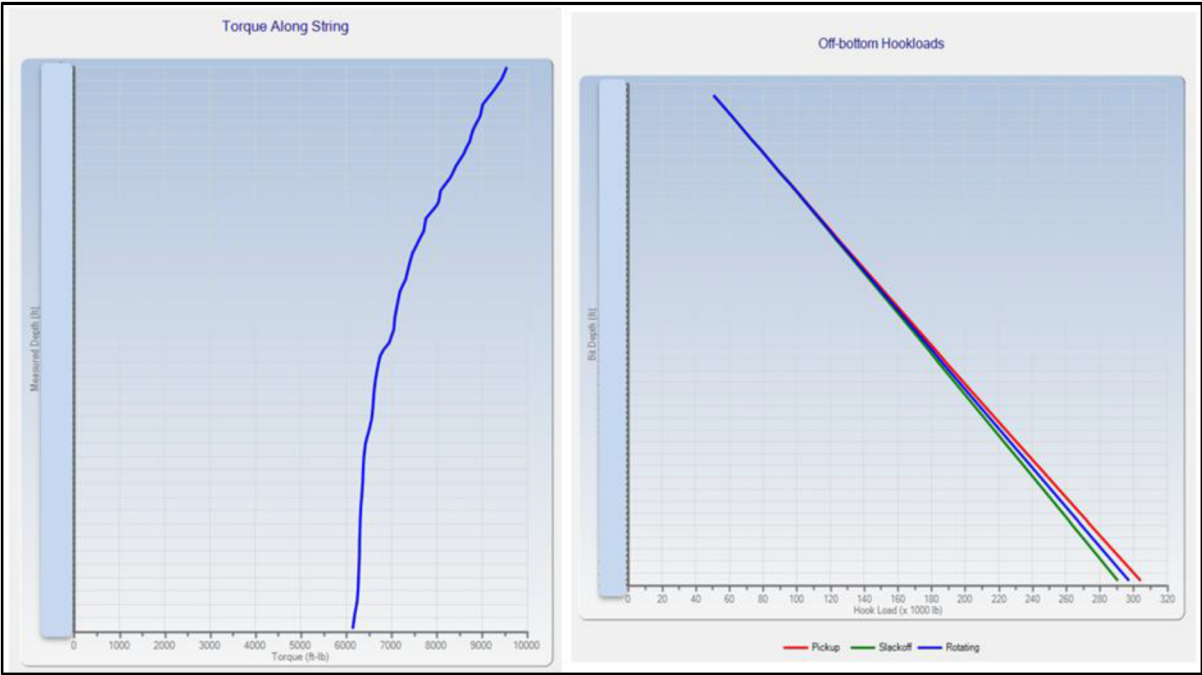


Figure 13—Planned T&D Graphs Simulation

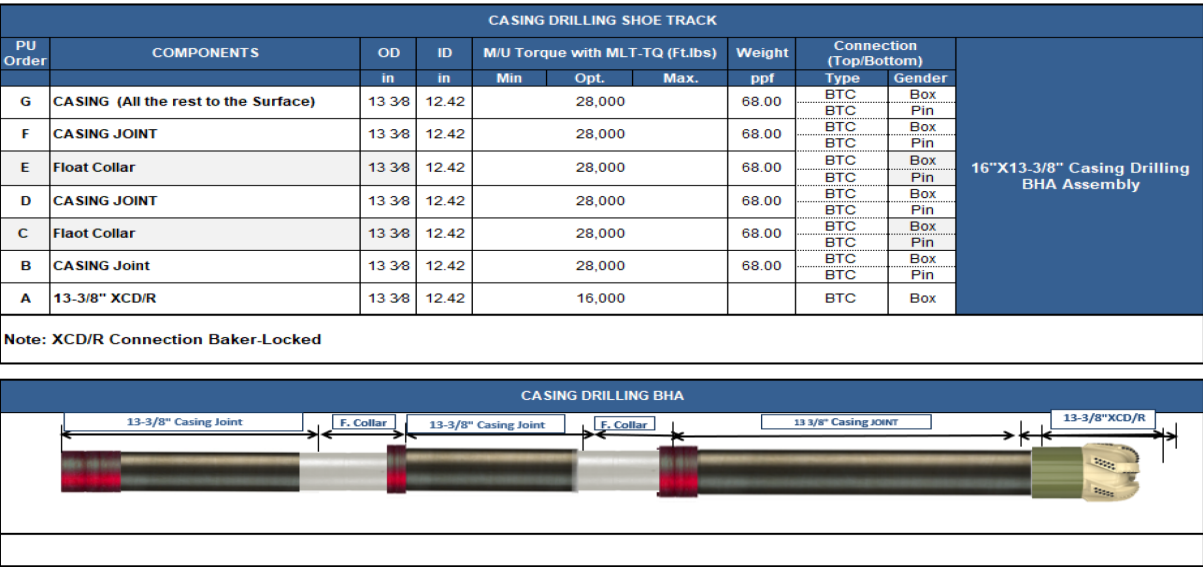


Figure 14—13-3/8-in level I casing-while-drilling (CwD) BHA

During job execution, the performance (Fig.15) was constantly monitored. The engineering simulations were updated using the actual parameters to ensure accurate measurements of the T&D furthermore,

the hydraulics are optimized real-time to ensure hole cleaning without further increase in the equivalent circulating density

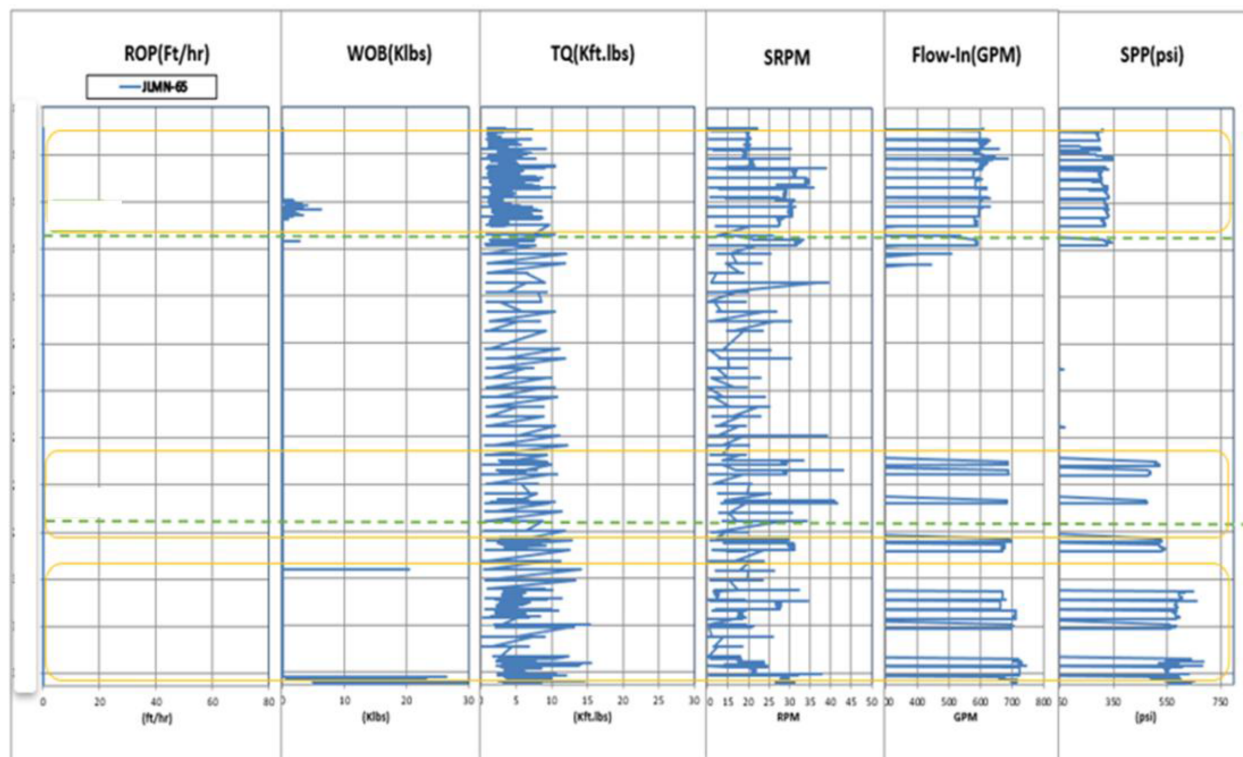


Figure 15—13-3/8-in Performance Analysis CwD

Conclusion & Future Considerations

The CwD technology has many benefits, such as eliminating risks related to instability in the top-hole sections and may reduce drilling cost by saving time. The initial startup and planning required a thorough engineering analysis performed by a multidisciplinary team. This is an ongoing design and engineering process and, after performing this job, it shows that further performance improvement and optimization can be achieved. Implementing CwD Level-I as a risk mitigation technology tool will be more beneficial & add extra value to enhance well delivery time & eliminate NPT related wellbore instability rather than applying it as a remedial after fact tool.

The most important achievement was introducing CwD as a solution for severe hole problems that resulted in suspending offset well. Even with the small number of incidents, hole troubles and stuck pipe can occur while drilling/tripping through severe unstable zones. Finding new approaches to optimize the performance and minimize the NPT when such troubles are experienced was the focus of the engineering team. After the successful global utilization of the technology, further plans are developed to encounter and build through such incidents for further optimization towards minimizing the NPT.

Abbreviations

BHA	: Bottomhole Assembly
BTC	: Buttress Thread Connection
CwD	: Casing-while-Drilling
DV	: Diverter Valve
MWD	: Measurement While Drilling
NPT	: Non-Productive Time

PDC : Polycrystalline Diamond Compact
Psi : Pounds per Square Inch
ROP : Rate of Penetration
RPM : Revolutions Per Minute
RSS : Rotary Steerable System
SPP : Stand-Pipe Pressure
TD : Target Depth
TQ : Torque
T&D : Torque and Drag
WOB : Weight-on-bit

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